

# *SyriaTel Customer Churn Analysis*

Predicting Customer Churn to Reduce Revenue Loss

# *Overview*

- SyriaTel is losing money every time a customer leaves
- This analysis predicts which customers are likely to churn
- Early identification allows the retention team to intervene

# *The Business Problem*

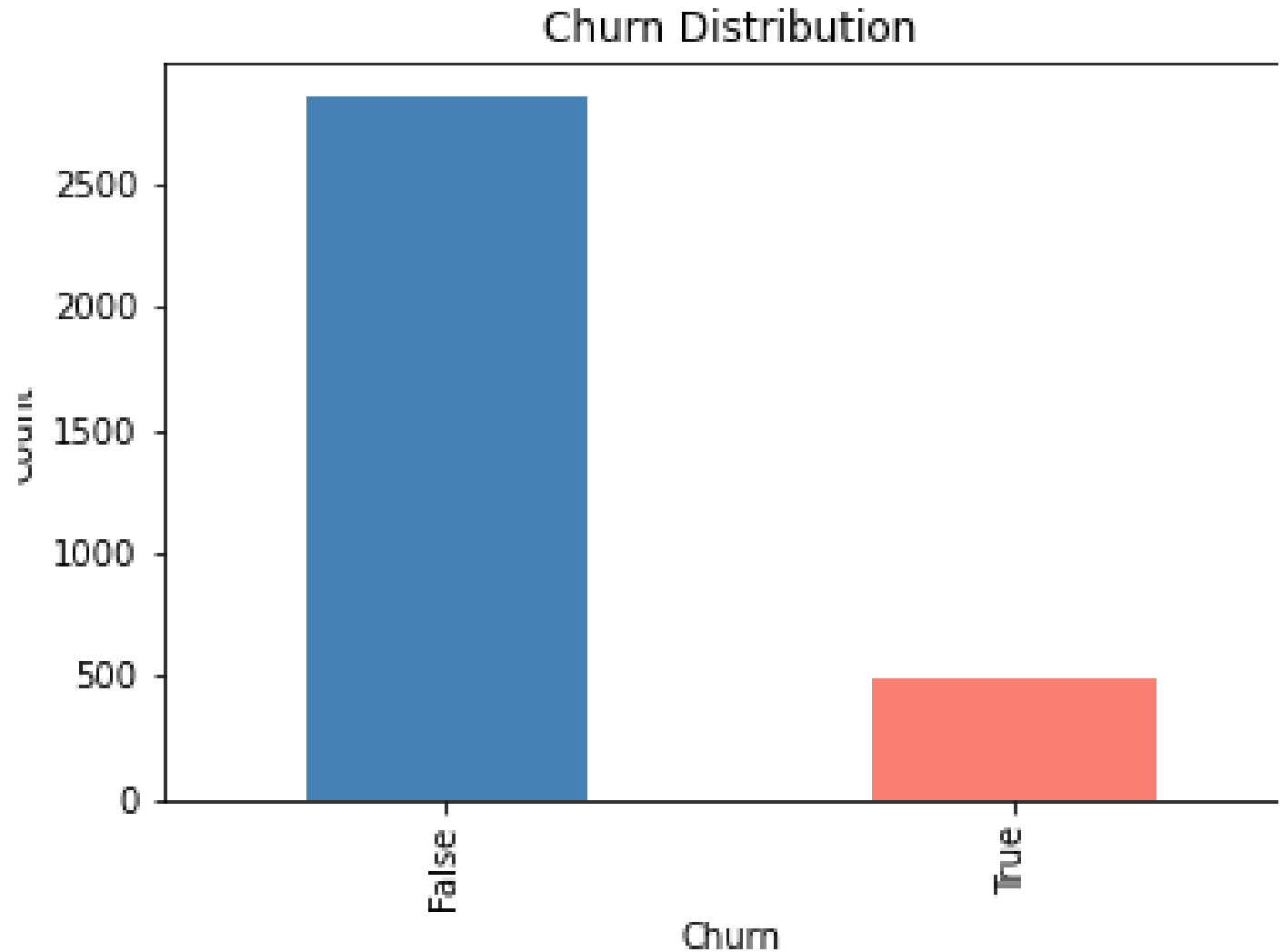
- Customer churn = lost revenue
- Key question: Are there predictable patterns in customer behavior?

# *About the Data*

- There are 3,333 customer records
- 21 features including call usage, charges and plans
- Only 14.5% of customers churned – imbalanced dataset

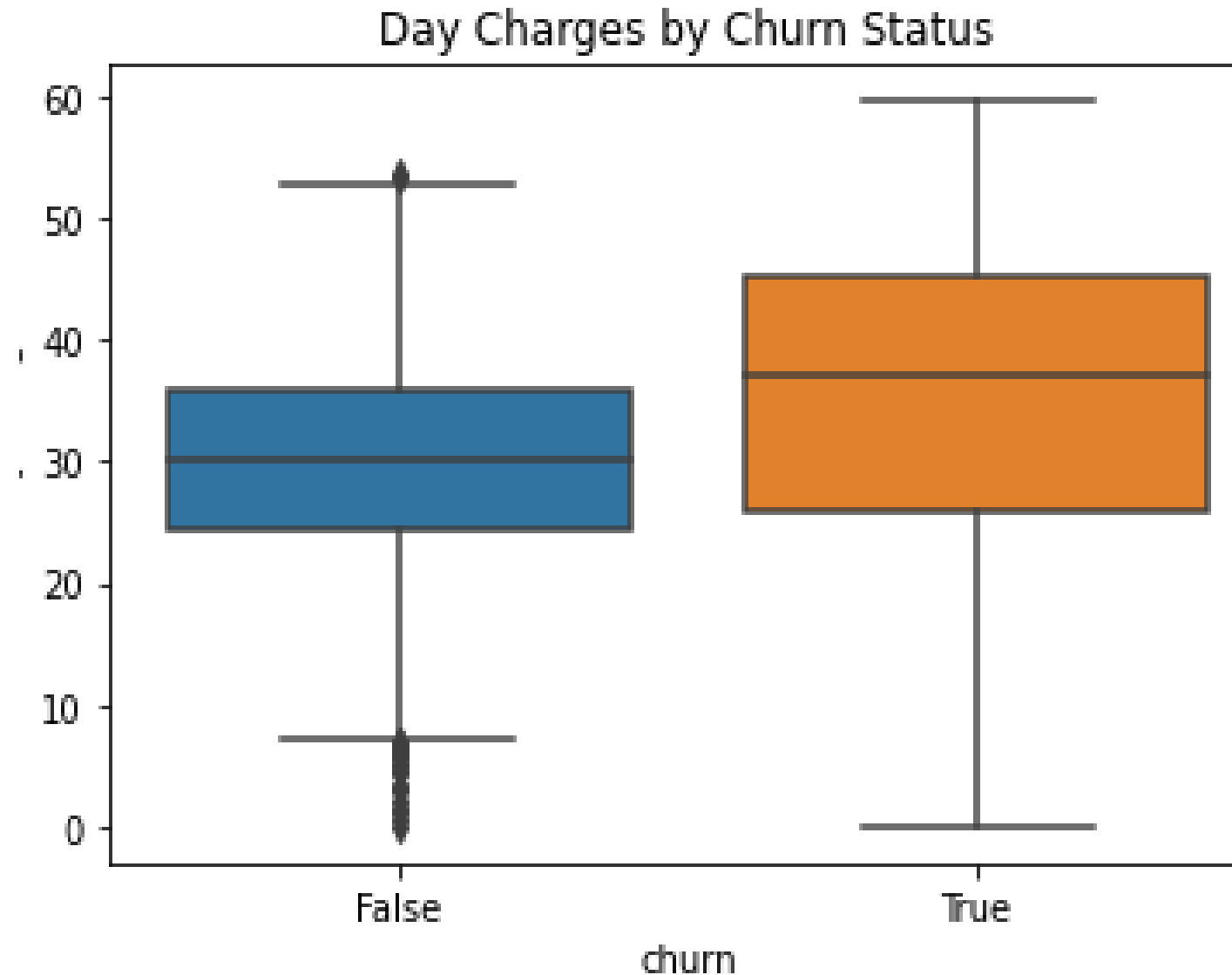
## *Churn Distribution*

Only 14.5% of customers churned (483 out of 3,333) — early identification is key to reducing revenue loss



## *What Drives Churn ?*

Customers who churned paid significantly higher day charges — targeting these customers early can prevent revenue loss



# *Our Approach*

- Built 3 models iteratively
- Model 1: Logistic Regression(baseline)
- Model 2: Decision Tree
- Model 3 : Tuned Decision Tree (final model)

## *How Well Does It Work?*

- Our model correctly identifies 94% of cases
- When it flags a churner, it is right 88% of the time
- Model performs consistently on both training and test data

# *Recommendations*

- Deploy the model to flag at-risk customers automatically
- Focus retention efforts on high day charge customers
- Offer discounts or upgrades to flagged customers

## *Next Steps*

- Try Random Forest for better performance
- Address class imbalance using SMOTE
- Monitor model performance over time

*Thank you*